SSC · GENERAL MATH · FINAL REVISION

Everything you need for today.

All formulas, all theorems, MCQ traps from 280 Final Shot slides, CQ patterns, and a 20 min phone sprint.

Algebra	Geometry	Trigonometry	Mensuration	Statistics
Ch 1–5, 11–13	Ch 6–8, 14–15	Ch 9–10	Ch 16	Ch 17

Today's plan (wake 8:00, stop 12:30, exam 2:00)

8:00–8:15	Wake, wash, eat something. No phone scrolling.
8:15–9:05	**Algebra core.** Ch 3 formulas (CQ), Ch 2 sets, then Ch 4 practice set (page "Ch 4"). 15 min max on practice.
9:05–9:50	**Theorems 17 to 38.** Read the proof format page first, then each theorem + one-line proof idea.
9:50–10:30	**Trig.** Ch 9 table + identities, Ch 10 all 8 scenarios.
10:30–11:15	**Ch 16 + Ch 17.** Formulas, then the "change" problems, then median/mode formula.
11:15–12:00	**MCQ-only chapters.** Ch 1, 5, 11, 12, 13, 14, 15 trap boxes. Skim, don't solve.
12:00–12:30	Buffer. If time left: re-read every red **Trap** box.
Commute	Open the **20 min Mobile Sprint** (last pages / separate phone PDF).

- **1** Real numbers
- **2** Set & function
- **3** Algebraic expressions
- **4** Exponent & log + practice
- **5** Equations (one variable)
- **11** Ratio & proportion
- **12** Simultaneous equations
- **13** Finite series
- ★ Theorem/construction answer format
- **6** Lines, angles, triangles
- **7** Practical geometry
- **8** Circle (Th 17–27)
- **14** Similarity, symmetry (Th 28–35)
- **15** Area theorems (Th 36–38)
- **9** Trig ratios
- **10** Height & distance
- **16** Mensuration
- **17** Statistics
- ⚡ 20 min Mobile Sprint

Source note: MCQ traps come from your Final Shot slides. Formulas and theorem statements follow the NCTB General Math book. Your book PDFs were on your PC (E:\, X:\) and did not upload, so if any theorem number differs, trust your book.

1 বাস্তব সংখ্যা Real numbers

MCQ NOT READ

Number family

$\mathbb{N}$ (স্বাভাবিক) $= \{1, 2, 3, \dots\}$. 0 is **not** natural.

$\mathbb{Z}$ (পূর্ণসংখ্যা) $= \{\dots, -2, -1, 0, 1, 2, \dots\}$. 0 is an integer.

মূলদ (rational) $= \frac{p}{q}$, $p, q \in \mathbb{Z}$, $q \neq 0$.

অমূলদ (irrational): $\sqrt{2}$, $\sqrt{8}$, π. $\frac{22}{7}$ is rational.

Real = rational + irrational.

Decimal test

সসীম দশমিক (terminating) → rational.

অসীম আবৃত্ত (repeating) → rational.

অসীম অনাবৃত্ত (non-repeating) → irrational.

$\sqrt{n}$ is irrational if n is not a perfect square.

$\sqrt{0.0025} = 0.05$ → সসীম দশমিক.

Repeating decimal → fraction (আবৃত্ত দশমিককে ভগ্নাংশে)

$$\frac{(\text{whole number without point}) - (\text{non-repeating part})}{\text{one 9 per repeating digit, then one 0 per non-repeating decimal digit}}$$

$2.3\dot{4} = \frac{234 - 23}{90} = \frac{211}{90}$ · $1.\dot{4}\dot{3} = \frac{143 - 1}{99} = \frac{142}{99}$ · $0.2\dot{7} = \frac{27 - 2}{90} = \frac{5}{18}$ · $0.\dot{5} = \frac{5}{9}$

To add repeating decimals fast: change all to fractions, add, convert back. $0.2\dot{7} + 0.\dot{5} = \frac{5}{18} + \frac{5}{9} = \frac{5}{6} = 0.8\dot{3}$

MCQ TRAPS (FROM YOUR SLIDES)

- $\frac{\sqrt{27}}{\sqrt{48}} = \sqrt{\frac{9}{16}} = \frac{3}{4}$ is **rational**. Always simplify the fraction under the root first. $\frac{\sqrt{32}}{\sqrt{64}} = \sqrt{\frac{1}{2}}$ is irrational.
- Two irrationals: sum, difference, product can each be rational **or** irrational. ($\sqrt{2} + (2 - \sqrt{2}) = 2$, $\sqrt{2} \cdot \sqrt{2} = 2$). So "always irrational" is **false**.
- 4 consecutive naturals product $+1$ = perfect square. $2 \cdot 3 \cdot 4 \cdot 5 + 1 = 121 = 11^2$.
- $a < b$: multiply by $c > 0$ keeps sign, $c < 0$ flips sign. Adding any c keeps order.
- $\sqrt{81} = 9$ is odd. $0.21 = \frac{21}{100}$ is a প্রকৃত (proper) fraction. 0 is non-negative.

CQ (ক) FAVOURITE

Prove $\sqrt{2}$ (or $\sqrt{3}$) is irrational: assume $\sqrt{2} = \frac{p}{q}$ in lowest terms, square: $p^2 = 2q^2$, so p even, then q even too. Contradiction with lowest terms.

2 সেট ও ফাংশন Set & function

TOP PRIORITY CQ + MCQ

Counting

$n(A) = m \Rightarrow n(P(A)) = 2^m$

Proper subsets $= 2^m - 1$

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$n(A \times B) = n(A) \cdot n(B)$

$P(\varnothing) = \{\varnothing\}$ (has 1 element, not empty)

Operations

$A \cap B = \{x : x \in A \text{ and } x \in B\}$

$A \cup B = \{x : x \in A \text{ or } x \in B\}$

$A \setminus B$ = in A, not in B

$A' = A^c = U \setminus A$; $A \cup A' = U$, $A \cap A' = \varnothing$

$A \cap B = \varnothing \Rightarrow$ নিশ্ছেদ (disjoint)

Laws (CQ proofs)

De Morgan: $(A \cup B)' = A' \cap B',\ (A \cap B)' = A' \cup B'$

Distributive: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$A \times (B \cap C) = (A \times B) \cap (A \times C)$

$A \times (B \cup C) = (A \times B) \cup (A \times C)$

Relation & function

$(a, b) = (c, d) \Rightarrow a = c,\ b = d$

ডোমেন = all first entries. রেঞ্জ = all second entries.

Function: each x has exactly one y.

$f(a)$: put $x = a$ everywhere. $f(\frac{1}{x})$: put $\frac{1}{x}$ for x.

MCQ TRAPS

- Set-builder: solve the condition first. $\{x \in \mathbb{N} : x^2 - 6x + 8 = 0\} = \{2, 4\}$. $\{x \in \mathbb{Z} : x^2 < 13\} = \{-3, \ldots, 3\}$. $\{x \in \mathbb{R} : x^2 + 9 = 0\} = \varnothing$.
- Relation $S = \{(x, y) : x, y \in A,\ y = x^2\}$: keep only pairs where y is **also in A**. Then read domain/range.
- No prime between 23 and 29 → empty set. $\{y \in \mathbb{N} : y^2 < 100 < y^3\} = \{5, \ldots, 9\}$ is finite.
- $A \cap B = \varnothing$ only tells "disjoint". Finite/infinite is not decided.
- $f(x) = 0$ with sum of roots $a + \frac{1}{a}$, product 1 → roots a and $\frac{1}{a}$.

CQ PATTERN

(ক) list a set / find $P(A)$. (খ) prove De Morgan or $A \times (B \cap C)$ law with given sets (write both sides fully, show equal). (গ) Venn word problem (students like X, Y, both, neither): use $n(A \cup B)$ formula, neither = total $- n(A \cup B)$. Or: domain and range of relation, $f(x)$ value problems.

3 বীজগাণিতিক রাশি Algebraic expressions

TOP PRIORITY CQ SURE

Square formulas

$(a + b)^2 = a^2 + 2ab + b^2$

$(a - b)^2 = a^2 - 2ab + b^2$

$a^2 + b^2 = (a + b)^2 - 2ab = (a - b)^2 + 2ab$

$(a + b)^2 = (a - b)^2 + 4ab;\ (a - b)^2 = (a + b)^2 - 4ab$

$(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$

$(a + b)^2 - (a - b)^2 = 4ab$

$ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$

$a^2 - b^2 = (a + b)(a - b)$

$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

$a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca)$

$(x + a)(x + b) = x^2 + (a + b)x + ab$

Cube formulas

$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

$= a^3 + b^3 + 3ab(a + b)$

$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

$= a^3 - b^3 - 3ab(a - b)$

$a^3 + b^3 = (a + b)(a^2 - ab + b^2) = (a + b)^3 - 3ab(a + b)$

$a^3 - b^3 = (a - b)(a^2 + ab + b^2) = (a - b)^3 + 3ab(a - b)$

$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

$= \frac{1}{2}(a + b + c)\{(a - b)^2 + (b - c)^2 + (c - a)^2\}$

If $a + b + c = 0$ **then** $a^3 + b^3 + c^3 = 3abc$

$x \pm \frac{1}{x}$ family (huge in MCQ)

$x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2 = \left(x - \frac{1}{x}\right)^2 + 2$

$\left(x - \frac{1}{x}\right)^2 = \left(x + \frac{1}{x}\right)^2 - 4$

$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$

$x^3 - \frac{1}{x^3} = \left(x - \frac{1}{x}\right)^3 + 3\left(x - \frac{1}{x}\right)$

$x + \frac{1}{x} = 2 \Rightarrow x = 1 \Rightarrow x^n + \frac{1}{x^n} = 2$ for any n

$x^4 - 5x^2 + 1 = 0$: divide by $x^2 \Rightarrow x^2 + \frac{1}{x^2} = 5$

Remainder & factor theorem

$f(x) \div (x - a)$: remainder $= f(a)$

$f(x) \div (ax - b)$: remainder $= f\left(\frac{b}{a}\right)$

$f(a) = 0 \Rightarrow (x - a)$ is a factor

Sum of coefficients $= 0 \Rightarrow (x - 1)$ is a factor

Middle term split: $x^2 + px + q$, find two numbers with sum p, product q

Surds

$x = \sqrt{5} + \sqrt{3} \Rightarrow \frac{1}{x} = \frac{\sqrt{5} - \sqrt{3}}{2}$ (multiply by conjugate)

$P = 3 + 2\sqrt{2} \Rightarrow \frac{1}{P} = 3 - 2\sqrt{2}$, $P + \frac{1}{P} = 6$

$5 + 2\sqrt{6} = (\sqrt{3} + \sqrt{2})^2$

Real-life formulas (বাস্তব সমস্যা)

Simple interest $I = \dfrac{Pnr}{100}$ (or Pnr with r as decimal)

Amount $A = P + I = P(1 + nr) \Rightarrow P = \dfrac{A}{1 + nr}$

Compound amount $C = P(1 + r)^n$

Compound interest $= P(1 + r)^n - P$

Growth: $P(1 + r)^n$ · Decline: $P(1 - r)^n$

Profit: $SP = CP\left(1 + \frac{x}{100}\right)$. Loss: $SP = CP\left(1 - \frac{x}{100}\right)$

25% profit $\Rightarrow CP : SP = 100 : 125 = 4 : 5$. 20% loss $\Rightarrow 5 : 4$

Boat: down $= u + v$, up $= u - v$. Stream $= \frac{\text{down} - \text{up}}{2}$

Work: A in a days $\Rightarrow$ 1 day $= \frac{1}{a}$ of work. Together $\frac{1}{a} + \frac{1}{b}$

Distance $=$ speed $\times$ time

CQ PATTERN

(ক) factorize or find $\frac{1}{x}$. (খ) given $x = \sqrt{a} + \sqrt{b}$, find $x^3 + \frac{1}{x^3}$ or $x^2 + \frac{1}{x^2}$: find $\frac{1}{x}$ first, then $x + \frac{1}{x}$, then use the family formulas. (গ) prove $a^3 + b^3 + c^3 = 3abc$ when $a + b + c = 0$, or simplify using $a^3 + b^3 + c^3 - 3abc$. Interest CQ: compare simple vs compound for same P, r, n.

4 সূচক ও লগারিদম Exponent & log

IMPORTANT NOT READ PRACTICE BELOW

Exponent laws ($a > 0$)

$a^m \cdot a^n = a^{m+n}$ · $\dfrac{a^m}{a^n} = a^{m-n}$

$(a^m)^n = a^{mn}$ · $(ab)^n = a^n b^n$ · $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

$a^0 = 1$ **(only if** $a \neq 0$**)** · $a^{-n} = \dfrac{1}{a^n}$

$a^{1/n} = \sqrt[n]{a}$ · $a^{m/n} = \sqrt[n]{a^m}$

$a^x = a^y \Rightarrow x = y$ $(a \neq 1)$ · $a^x = b^x \Rightarrow a = b$

$(x^{-1} + y^{-1})^{-1} = \dfrac{xy}{x + y}$

Log laws ($a > 0,\ a \neq 1,\ M, N > 0$)

$a^x = N \iff \log_a N = x$

$\log_a 1 = 0$ · $\log_a a = 1$ · $a^{\log_a N} = N$

$\log_a(MN) = \log_a M + \log_a N$

$\log_a \frac{M}{N} = \log_a M - \log_a N$

$\log_a M^r = r \log_a M$

Base change: $\log_a M = \dfrac{\log_b M}{\log_b a}$; $\log_a b \cdot \log_b a = 1$

$\log_{a^k} N = \frac{1}{k} \log_a N$ (e.g. $\log_{49} 7 = \frac{1}{2}$, $\log_{\sqrt{7}} 7 = 2$)

Log of 0 or a negative number: **not defined**

Scientific form, পূর্ণক (characteristic), অংশক (mantissa)

Scientific form: $a \times 10^n$, $1 \le a < 10$. $0.00052 = 5.2 \times 10^{-4}$

Characteristic $= n$ (the power of 10)

Number ≥ 1: characteristic $=$ (digits before point) -1. $\log 62.542 \to 1$

Number < 1: negative, written with bar. $0.00000538 = 5.38 \times 10^{-6} \to \bar{6}$

Mantissa is **always positive** (from table). $\log 0.00234$: mantissa 0.3692

$\log 0.00234 = \bar{3}.3692 = -2.6308$

MCQ TRAPS

- $\log_x 9 = 2 \Rightarrow x = 3$ only (base can't be -3).
- $\dfrac{16^a - 1}{4^a - 1} = \dfrac{(4^a)^2 - 1}{4^a - 1} = 4^a + 1 = 2^{2a} + 1$.
- $\log_5(\sqrt[3]{5} \cdot \sqrt{5}) = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$. $\log_{2\sqrt{5}} 40\sqrt{5} = 3$ since $(2\sqrt{5})^3 = 40\sqrt{5}$.
- $2\log x - \log(2x - 1) = 0 \Rightarrow x^2 = 2x - 1 \Rightarrow x = 1$. Always check $x > 0$ in logs.
- $16^{x+1} = 4^{x-2}$: make same base: $4^{2x+2} = 4^{x-2} \Rightarrow x = -4$.

PRACTICE: 2 SETS, ABOUT 15 MIN TOTAL

Set A **Quick log drills** (do all 6, about 6 min)

1. $\log_{2\sqrt{3}} 144 = ?$
2. $\log_x 324 = 4$ হলে $x = ?$
3. প্রমাণ কর: $\log \dfrac{a^2}{bc} + \log \dfrac{b^2}{ca} + \log \dfrac{c^2}{ab} = 0$
4. সমাধান কর: $\log_2(x + 3) + \log_2(x - 3) = 4$
5. $\log 2 = 0.3010,\ \log 3 = 0.4771$ হলে $\log 72 = ?$
6. 0.000325 এর সাধারণ লগের পূর্ণক কত?

Set B **Board-style CQ** (about 9 min)

উদ্দীপক: $a^x = b,\ b^y = c,\ c^z = a$

(ক) $\log_{\sqrt{3}} 81$ এর মান নির্ণয় কর। (খ) উদ্দীপক থেকে প্রমাণ কর যে, $xyz = 1$। (গ) সমাধান কর: $\log_{10}\left(98 + \sqrt{x^2 - 12x + 36}\right) = 2$

ANSWERS (CHECK AFTER, NOT BEFORE)

A1 $(2\sqrt{3})^2 = 12,\ 12^2 = 144 \Rightarrow$ **4**. **A2** $x^4 = 324 \Rightarrow x^2 = 18 \Rightarrow x = 3\sqrt{2}$. **A3** add logs: $\log \dfrac{a^2b^2c^2}{a^2b^2c^2} = \log 1 = 0$. **A4** $(x + 3)(x - 3) = 2^4 \Rightarrow x^2 = 25 \Rightarrow x = 5$ (reject -5, log of negative). **A5** $72 = 2^3 \cdot 3^2 \Rightarrow 0.9030 + 0.9542 = 1.8572$. **A6** $3.25 \times 10^{-4} \Rightarrow \bar{4}$.

B(ক) $(\sqrt{3})^8 = 3^4 = 81 \Rightarrow$ **8**. **B(খ)** take log: $x = \dfrac{\log b}{\log a}$, $y = \dfrac{\log c}{\log b}$, $z = \dfrac{\log a}{\log c}$. Multiply $\Rightarrow xyz = 1$. **B(গ)** $98 + \sqrt{(x - 6)^2} = 10^2 \Rightarrow |x - 6| = 2 \Rightarrow x = 8$ or $x = 4$. Both work.

5 এক চলকবিশিষ্ট সমীকরণ Equations in one variable

MCQ NOT READ

Must know (from zero)

সমীকরণ: true for some values only. **অভেদ** (identity): true for **all** values.

Every formula is an identity. Every identity is an equation. Not every equation is an identity.

ঘাত (degree) = highest power. Number of roots ≤ degree.

Answer as সমাধান সেট: $S = \{\dots\}$

$(x+1)^2 - (x-1)^2 = 4x$ is an identity.

Solving tools

Linear: move terms, divide. Fractions: cross-multiply.

Quadratic: factor ⇒ each factor $= 0$.

Or: $x = \dfrac{-b \pm \sqrt{b^2-4ac}}{2a}$

Root equation: isolate $\sqrt{}$, square, then **check every answer**.

Word problems: 2-digit number $= 10x + y$; digits swapped $= 10y + x$.

MCQ TRAPS

- $x^2 = 3x$: never divide by x! $x(x-3) = 0 \Rightarrow S = \{0, 3\}$. Same for $2x(x-3) = x - 3 \Rightarrow \{3, \frac{1}{2}\}$.
- $\sqrt{2x-5} + 3 = 2 \Rightarrow \sqrt{} = -1$, impossible ⇒ $S = \varnothing$.
- If answer makes a denominator 0, reject it. $\dfrac{P-2}{P-1} = 2 - \dfrac{1}{P-1}$ gives $P = 1 \Rightarrow S = \varnothing$.
- $(x^2-3)^2 = 0$ has 2 roots ($\pm\sqrt{3}$). $x^2 - \frac{4}{x^2} = 0 \Rightarrow x^4 - 4 = 0$: degree 4, constant term -4, real roots $\pm\sqrt{2}$.
- Tens digit = 3 × units digit $P \Rightarrow$ number $= 31P$.

11 বীজগাণিতিক অনুপাত ও সমানুপাত Ratio & proportion

CQ + MCQ NOT READ

Basics

$a : b = \dfrac{a}{b}$ (same units only)

সমানুপাত: $a : b = c : d \Rightarrow ad = bc$

ক্রমিক সমানুপাতী: $a : b = b : c \Rightarrow b^2 = ac$

মধ্যসমানুপাতী $b = \sqrt{ac}$. Example: $4, 6, 9$

If $\dfrac{a}{b} = \dfrac{c}{d} = \dfrac{e}{f} = k$, then each $= \dfrac{a+c+e}{b+d+f}$

The 5 properties (names matter)

Given $\dfrac{a}{b} = \dfrac{c}{d}$:

ব্যস্তকরণ: $\dfrac{b}{a} = \dfrac{d}{c}$

একান্তরকরণ: $\dfrac{a}{c} = \dfrac{b}{d}$

যোজন: $\dfrac{a+b}{b} = \dfrac{c+d}{d}$

বিয়োজন: $\dfrac{a-b}{b} = \dfrac{c-d}{d}$

যোজন-বিয়োজন: $\dfrac{a+b}{a-b} = \dfrac{c+d}{c-d}$

Biggest CQ pattern: যোজন-বিয়োজন twice

If $x = \dfrac{\sqrt{a+b} + \sqrt{a-b}}{\sqrt{a+b} - \sqrt{a-b}}$, show $bx^2 - 2ax + b = 0$.

Step 1: যোজন-বিয়োজন ⇒ $\dfrac{x+1}{x-1} = \dfrac{\sqrt{a+b}}{\sqrt{a-b}}$. Step 2: square ⇒ $\dfrac{(x+1)^2}{(x-1)^2} = \dfrac{a+b}{a-b}$.

Step 3: যোজন-বিয়োজন again ⇒ $\dfrac{2(x^2+1)}{4x} = \dfrac{2a}{2b} \Rightarrow b(x^2+1) = 2ax \Rightarrow bx^2 - 2ax + b = 0$. ✓

Same trick: $\dfrac{2a}{y} = \dfrac{m^2+n^2}{mn} \Rightarrow \dfrac{a-y}{a+y} = \left(\dfrac{m-n}{m+n}\right)^2$

Word problems

Join ratios: $a:b=3:4$, $b:c=6:7 \Rightarrow$ make b same (24) $\Rightarrow 9:12:14$

Ratio $3:4$, HCF 3 $\Rightarrow$ numbers 9, 12 $\Rightarrow$ LCM 36

Ratio $3:2$, LCM 42 $\Rightarrow 6k=42 \Rightarrow$ HCF $k=7$

Divide T in $a:b$: parts $\frac{a}{a+b}T$, $\frac{b}{a+b}T$

Length $+20\%$, width $-10\% \Rightarrow 1.2\times 0.9=1.08 \Rightarrow$ area $+8\%$

Side $-10\% \Rightarrow 0.9^2=0.81 \Rightarrow$ area -19%

Side ×3 $\Rightarrow$ area ×9 (increase = 8 times)

Partnership profit ratio = capital × time

MCQ TRAPS

- a,b,c ক্রমিক সমানুপাতী: $b^2=ac$ ✓, $\frac{a+b}{b}=\frac{b+c}{c}$ ✓, but $2b=a+c$ ✗ (that is AP).
- Circumference : radius $=2\pi:1$. Square area : square-on-diagonal area $=1:2$.
- Triangle angles $3:4:5 \Rightarrow 45^\circ, 60^\circ, 75^\circ$.

12 দুই চলকবিশিষ্ট সরল সহসমীকরণ Simultaneous equations

MCQ + CQ · NOT READ

Methods

প্রতিস্থাপন (substitution): get y from one, put in other.

অপনয়ন (elimination): make one coefficient equal, add/subtract.

আড়গুণন (cross-multiplication) for $a_1x+b_1y+c_1=0$, $a_2x+b_2y+c_2=0$:

$$\frac{x}{b_1c_2-b_2c_1}=\frac{y}{c_1a_2-c_2a_1}=\frac{1}{a_1b_2-a_2b_1}$$

Graph: find 3 points each line, the meeting point is the answer.

Consistency test (MCQ favourite)

Condition	Type	Solutions
$\frac{a_1}{a_2}\neq\frac{b_1}{b_2}$	সমঞ্জস, অনির্ভরশীল	1 (lines cross)
$\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$	সমঞ্জস, নির্ভরশীল	Infinite (same line)
$\frac{a_1}{a_2}=\frac{b_1}{b_2}\neq\frac{c_1}{c_2}$	অসমঞ্জস	None (parallel)

$3x-5y=7$, $6x-10y=15$: $\frac{1}{2}=\frac{1}{2}\neq\frac{7}{15} \Rightarrow$ no solution.

CQ PATTERN

(ক) check consistency. (খ) solve by cross-multiplication. (গ) graph both lines and read the meeting point. Word problems: fraction $\frac{x}{y}$ ("add 1 to numerator, it becomes 1": $\frac{x+1}{y}=1$), digits, ages, rectangle sides. Check a point on a line by plugging in.

13 সসীম ধারা Finite series

CQ + MCQ

Arithmetic (সমান্তর), first term a, difference d

$d=t_2-t_1$

nth term $t_n=a+(n-1)d$

$S_n=\frac{n}{2}\{2a+(n-1)d\}=\frac{n}{2}(a+l)$

Number of terms $n=\frac{l-a}{d}+1$

$t_n=S_n-S_{n-1}$

Geometric (গুণোত্তর), ratio r

$r=\frac{t_2}{t_1}$

$t_n=ar^{n-1}$

$S_n=\frac{a(r^n-1)}{r-1}$ $(r>1)$

$S_n=\frac{a(1-r^n)}{1-r}$ $(r<1)$

$r=-1$: $2-2+2-\ldots \Rightarrow$ even term $=-2$, sum of $2n$ terms $=0$

Natural number sums

$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$

$1^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$

$1^3 + \cdots + n^3 = \left\{\frac{n(n+1)}{2}\right\}^2$

First n odd numbers $= n^2$

First n even numbers $= n(n+1)$

$2 + 3 + \cdots + 50 = 1275 - 1 = 1274$

MCQ TRAPS

- $\log 2 + \log 4 + \log 8 + \ldots$ is an AP in $\log 2$: 8th term $= \log 2^8 = \log 256$.
- $7, x, y, 56$ GP $\Rightarrow 7r^3 = 56 \Rightarrow r = 2, y = 28$. $1, \frac{1}{3}, \frac{1}{7}, \frac{1}{15} \Rightarrow \frac{1}{2^n - 1}$.
- Sum of cubes is $\frac{n^2(n+1)^2}{4}$, not $\frac{n^3(n+1)^3}{8}$.

How to write a theorem / construction উপপাদ্য ও সম্পাদ্যের ফরম্যাট

FREE MARKS

উপপাদ্য (theorem) – 4 parts

1. সাধারণ নির্বচন – write the statement in words (copy the theorem line).

2. চিত্র – draw a clean labeled figure.

3. বিশেষ নির্বচন – "মনে করি, ABC একটি ত্রিভুজ ... প্রমাণ করতে হবে যে, ..." (use figure letters).

4. অঙ্কন – extra lines you draw ("O, A যোগ করি"). Skip if none.

5. প্রমাণ – table with two columns: ধাপ (step) | যথার্থতা (reason). End with "[প্রমাণিত]".

সম্পাদ্য (construction) – parts

1. বিশেষ নির্বচন – "মনে করি, a, b দুইটি রেখাংশ ও ∠x একটি কোণ। এমন একটি ত্রিভুজ আঁকতে হবে যেন ..."

2. Given data figure – draw the given lines/angles first, labeled.

3. অঙ্কন – numbered steps: "যেকোনো রশ্মি BE থেকে BC = a কেটে নিই..."

4. প্রমাণ – short: show the figure has the given sides/angles.

Keep construction arcs visible. Never erase them. Use a sharp pencil.

REASONS YOU WILL WRITE AGAIN AND AGAIN (যথার্থতা)

একই বৃত্তের ব্যাসার্ধ (radii equal) · সাধারণ বাহু (common side) · বিপ্রতীপ কোণ (vertical angles) · একান্তর কোণ (alternate angles, parallel lines) · অনুরূপ কোণ · বাহু-কোণ-বাহু / বাহু-বাহু-বাহু / কোণ-বাহু-কোণ / অতিভুজ-বাহু উপপাদ্য (SAS/SSS/ASA/RHS) · সমদ্বিবাহু ত্রিভুজের ভূমিসংলগ্ন কোণ · ত্রিভুজের বহিঃস্থ কোণ · সরলকোণ = 180°.

GEOMETRY CQ SHAPE

(ক) small fact or a figure from data (2 marks). (খ) prove a book theorem, often with new letters (4 marks). (গ) prove a related result or do a construction (4 marks). If you recognise the theorem, just rename letters and write the book proof.

6 রেখা, কোণ ও ত্রিভুজ Lines, angles, triangles

MCQ 50/50

Angles

পূরক (complementary) sum 90°. সম্পূরক (supplementary) sum 180°.

বিপ্রতীপ কোণ (vertical) are equal.

Parallel lines + transversal: alternate equal, corresponding equal, same-side interior sum 180°.

Surface boundary = line. Line boundary = point. Point has no length, width, height.

Triangle facts

Angle sum 180°. Exterior angle $=$ sum of two opposite interior angles.

Sum of any two sides $>$ third. Difference $<$ third.

Bigger side faces bigger angle (and back).

Isosceles: equal sides $\Leftrightarrow$ equal base angles.

Midpoint theorem: joining midpoints of 2 sides $\parallel$ third and $= \frac{1}{2}$ of it.

Right triangle 30°–60°–90°: side opposite $30^\circ = \frac{1}{2}$ hypotenuse.

Congruence (সর্বসমতা)

SAS, SSS, ASA (or AAS), RHS. **SSA and AAA do not prove congruence.**

Centres

ভরকেন্দ্র (centroid): medians meet, divides each $2:1$.

পরিকেন্দ্র (circumcentre): perpendicular bisectors meet. Acute: inside. Right: midpoint of hypotenuse. Obtuse: outside.

অন্তঃকেন্দ্র (incentre): angle bisectors meet, always inside.

লম্ববিন্দু (orthocentre): altitudes meet.

Shortcuts

Bisectors of $\angle B$, $\angle C$ meet at O $\Rightarrow \angle BOC = 90^\circ + \frac{\angle A}{2}$

Polygon interior sum $= (n-2) \times 180^\circ$

Regular polygon: each exterior $= \frac{360^\circ}{n}$, interior $= \frac{(n-2)180^\circ}{n}$

Angles $1:1:2 \Rightarrow 45, 45, 90$: right isosceles.

$\angle A = \angle B + \angle C \Rightarrow \angle A = 90^\circ$

MCQ TRAPS

- Equilateral $\triangle$, BC extended to D: $\angle ACD = 120^\circ$, $\angle ABC + \angle ACD = 180^\circ$.
- $AC = BC$, $\angle A = 25^\circ \Rightarrow$ equal angles are A and $B \Rightarrow \angle C = 130^\circ$. Equal sides face equal angles; find which!
- Midpoint: $BC = 2EF$. If $EF = 3x + 5$, $BC = 12x - 8 \Rightarrow x = 3$, $BC = 28$.

7 ব্যবহারিক জ্যামিতি Practical geometry

MCQ · DRAWING FROM BOOK

Can it be drawn?

3 sides: only if each pair sum $>$ third. $6, 9, 13$ ✓ $4, 7, 11$ ✗ (equal, flat).

Sides 4 and 10 ⇒ third side between 6 and 14 (e.g. 8).

3 angles: must sum to 180°, and still don't fix size.

Data needed (উপাত্ত)

Shape	Min data
Quadrilateral (general)	5
Trapezium	4
Parallelogram	3 (2 adjacent sides + included angle)
Rectangle / Rhombus	2
Square	1 (side or perimeter)

Rhombus: side + angle, side + diagonal, or 2 diagonals. One side alone: ✗. Perimeter + one angle ⇒ rhombus.

8 বৃত্ত Circle

CQ LIKELY · EXERCISE NOT READ

Words

জ্যা chord · ব্যাস diameter (longest chord) · চাপ arc

উপচাপ minor arc · অধিচাপ major arc

কেন্দ্রস্থ কোণ at centre · বৃত্তস্থ কোণ on circle

স্পর্শক tangent · ছেদক secant · বৃত্তস্থ চতুর্ভুজ cyclic quad

Key numbers

Half chord $= \sqrt{r^2 - d^2}$ (d = distance from centre)

Inscribed angle $= \frac{1}{2}$ central angle (same arc)

Angle in semicircle $= 90^\circ$

Cyclic quad: opposite angles sum 180°

Tangent length from P: $\sqrt{OP^2 - r^2}$

Touch externally: $d = r_1 + r_2$. Internally: $d = |r_1 - r_2|$

Counting tangents

Point / circles	Point inside	On circle	Outside	Separate circles	Touch outside	Intersect	Touch inside	One inside other
Tangents	0	1	2	4 (2 direct + 2 তির্যক)	3	2	1	0

MCQ TRAPS

- Angle in অধিচাপ (major segment) is acute. In উপচাপ (minor segment) is obtuse. Inscribed 120° ⇒ vertex on minor arc.
- Point on minor-arc side: $\angle = \frac{360^\circ - \text{central}}{2}$. $\angle POQ = 88^\circ \Rightarrow 136^\circ$.
- Parallelogram inside a circle is a **rectangle**. Every triangle has 3 বহির্বৃত্ত (excircles).
- $r = 5$, chord distance 3 ⇒ half chord 4 ⇒ chord 8. $OA = 4$, $OD = 3 \Rightarrow AB = 2\sqrt{7}$.

THEOREMS 17–27 (CIRCLE), STATEMENT + ONE-LINE PROOF IDEA

17 **বৃত্তের কেন্দ্র ও ব্যাস ভিন্ন কোনো জ্যা-এর মধ্যবিন্দুর সংযোজক রেখাংশ ঐ জ্যা-এর উপর লম্ব।**

Join O to A, B. $\triangle OAM \cong \triangle OBM$ (SSS). The two angles at M are equal and sum to 180°, so each is 90°.

18 **বৃত্তের সকল সমান জ্যা কেন্দ্র থেকে সমদূরবর্তী।**

Drop $OE \perp AB$, $OF \perp CD$ (they bisect chords). Join OA, OC. $\triangle OAE \cong \triangle OCF$ (RHS: OA = OC, AE = CF). So OE = OF.

19 **বৃত্তের কেন্দ্র থেকে সমদূরবর্তী সকল জ্যা পরস্পর সমান।**

Converse of 18. OE = OF, OA = OC $\Rightarrow$ RHS $\Rightarrow$ AE = CF $\Rightarrow$ AB = 2AE = 2CF = CD.

20 **বৃত্তের একই চাপের উপর দণ্ডায়মান বৃত্তস্থ কোণ কেন্দ্রস্থ কোণের অর্ধেক।**

Join A (on circle) to O, extend to E. In isosceles $\triangle OAB$, exterior $\angle BOE = 2\angle BAO$. Same for C side. Add (or subtract).

21 **বৃত্তের একই চাপের উপর দণ্ডায়মান বৃত্তস্থ কোণগুলো পরস্পর সমান।**

Each one is half of the same central angle (Th 20). So all are equal. Corollary: angle in semicircle = 90°.

22 **বৃত্তে অন্তর্লিখিত চতুর্ভুজের যেকোনো দুইটি বিপরীত কোণের সমষ্টি দুই সমকোণ।**

Join O to B, D. The two central angles on BD sum to 360°. $\angle A$ and $\angle C$ are halves (Th 20) $\Rightarrow$ sum 180°.

23 **কোনো চতুর্ভুজের বিপরীত কোণদ্বয় সম্পূরক হলে তার শীর্ষবিন্দু চারটি সমবৃত্ত হয়।**

Converse of 22. Draw circle through A, B, C. Assume D not on it, meets at E. Then $\angle AEC = \angle ADC$, impossible (exterior angle). Contradiction.

24 **বৃত্তের যেকোনো বিন্দুতে অঙ্কিত স্পর্শক স্পর্শবিন্দুগামী ব্যাসার্ধের উপর লম্ব।**

Every other point Q on the tangent is outside the circle, so OQ > OP. Shortest distance is the perpendicular $\Rightarrow OP \perp$ tangent.

25 **বৃত্তের বহিঃস্থ কোনো বিন্দু থেকে বৃত্তে দুইটি স্পর্শক টানলে, ঐ বিন্দু থেকে স্পর্শবিন্দুদ্বয়ের দূরত্ব সমান।**

Join OA, OB, OP. $\angle OAP = \angle OBP = 90°$ (Th 24). OA = OB, OP common $\Rightarrow$ RHS $\Rightarrow$ PA = PB. (Also OP bisects $\angle APB$.)

26 **দুইটি বৃত্ত পরস্পর বহিঃস্পর্শ করলে, স্পর্শবিন্দু ও কেন্দ্রদ্বয় সমরেখ।**

Draw the common tangent at the touch point P. Both radii $\perp$ that tangent at P (Th 24). 90° + 90° = 180° $\Rightarrow$ straight line.

27 **দুইটি বৃত্ত পরস্পর অন্তঃস্পর্শ করলে, স্পর্শবিন্দু ও কেন্দ্রদ্বয় সমরেখ।**

Same idea: both radii $\perp$ common tangent at P, so both lie on one line through P.

14 অনুপাত, সদৃশতা ও প্রতিসমতা Ratio, similarity, symmetry

MCQ SKIPPED

Similarity (সদৃশতা)

Similar = same shape. For triangles: equal angles $\Leftrightarrow$ proportional sides.

$$\triangle ABC \sim \triangle DEF \Rightarrow \frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$

Area ratio $=$ (side ratio)^2. Sides $4:5 \Rightarrow$ areas $16:25$.

Heights, medians, perimeters: same ratio as sides.

$$DE \parallel BC \Rightarrow \frac{AD}{DB} = \frac{AE}{EC} \text{ and } \triangle ADE \sim \triangle ABC$$

Angle bisector AD $\Rightarrow \frac{BD}{DC} = \frac{AB}{AC}$

Symmetry (প্রতিসমতা)

Shape	Lines	Rotation order
Scalene triangle	0	1
Isosceles triangle	1	1
Equilateral triangle	3	3
Rectangle / Rhombus	2	2
Square	4	4
Regular n-gon	n	n
Parallelogram	0	2
Circle	infinite	infinite

Rotation angle $= \frac{360^\circ}{\text{order}}$. S, Z, H, N: order 2, angle 180°. T: 1 line. H: 2 lines.

THEOREMS 28–35 (RATIO & SIMILARITY)

28 **ত্রিভুজের যেকোনো বাহুর সমান্তরাল সরলরেখা ঐ ত্রিভুজের অপর বাহুদ্বয়কে বা এদের বর্ধিতাংশদ্বয়কে সমান অনুপাতে বিভক্ত করে।**

DE ∥ BC. Join BE, CD. △BDE = △CDE (same base DE, same parallels). △ADE/△BDE = AD/DB, △ADE/△CDE = AE/EC ⇒ AD/DB = AE/EC.

29 **কোনো সরলরেখা ত্রিভুজের দুই বাহুকে সমান অনুপাতে বিভক্ত করলে, ঐ সরলরেখা তৃতীয় বাহুর সমান্তরাল।**

Converse of 28, using the same area-ratio steps backwards: △BDE = △CDE on same base ⇒ same parallels ⇒ DE ∥ BC.

30 **ত্রিভুজের যেকোনো কোণের সমদ্বিখণ্ডক বিপরীত বাহুকে ঐ কোণ সংলগ্ন বাহুদ্বয়ের অনুপাতে অন্তর্বিভক্ত করে।**

AD bisects ∠A. Draw CE ∥ DA meeting BA extended at E. Then ∠ACE = ∠AEC ⇒ AE = AC. By Th 28: BD/DC = BA/AE = AB/AC.

31 **ত্রিভুজের যেকোনো বাহু অপর দুই বাহুর অনুপাতে অন্তর্বিভক্ত হলে, বিভাগ বিন্দু থেকে বিপরীত শীর্ষ পর্যন্ত অঙ্কিত রেখাংশ ঐ শীর্ষকোণের সমদ্বিখণ্ডক।**

Converse of 30. Same CE ∥ DA construction, then AE = AC ⇒ base angles equal ⇒ ∠BAD = ∠DAC.

32 **দুইটি ত্রিভুজ সদৃশকোণী হলে, তাদের অনুরূপ বাহুগুলো সমানুপাতিক।**

Cut AP = DE on AB, AQ = DF on AC. △APQ ≅ △DEF (SAS). ∠APQ = ∠E = ∠B ⇒ PQ ∥ BC ⇒ Th 28 gives the ratios.

33 **দুইটি ত্রিভুজের বাহুগুলো সমানুপাতিক হলে, অনুরূপ বাহুর বিপরীত কোণগুলো পরস্পর সমান।**

Converse of 32. Build a triangle on EF with angles of △ABC, it is similar (Th 32), its sides equal △DEF's ⇒ SSS ⇒ angles equal.

34 **দুইটি ত্রিভুজের একটির এক কোণ অপরটির এক কোণের সমান হলে এবং সমান সমান কোণ সংলগ্ন বাহুগুলো সমানুপাতিক হলে, ত্রিভুজদ্বয় সদৃশ।**

SAS similarity. Cut AP = DE, AQ = DF; △APQ ≅ △DEF; ratios ⇒ PQ ∥ BC (Th 29) ⇒ equal angles.

35 **দুইটি সদৃশ ত্রিভুজক্ষেত্রের ক্ষেত্রফলদ্বয়ের অনুপাত তাদের যেকোনো দুই অনুরূপ বাহুর উপর অঙ্কিত বর্গক্ষেত্রদ্বয়ের ক্ষেত্রফলের অনুপাতের সমান।**

Draw heights AP, DQ. △ABP ~ △DEQ ⇒ AP/DQ = AB/DE = BC/EF. Area ratio = (BC·AP)/(EF·DQ) = (BC/EF)^2.

15 ক্ষেত্রফল সম্পর্কিত উপপাদ্য ও সম্পাদ্য Area theorems

MCQ SKIPPED

Facts

Diagonal splits a parallelogram into 2 equal triangles: $\triangle ABD = \frac{1}{2}\square ABCD$.

A median splits a triangle into 2 equal-area triangles.

Same base + same parallels ⇒ equal area (triangles with triangles, parallelograms with parallelograms).

Square on the diagonal $= 2\times$ the square ⇒ square $= \frac{1}{2}$ of square-on-diagonal.

Pythagoras family

Right angle at C: $AB^2 = AC^2 + BC^2$

Converse: if $AB^2 = AC^2 + BC^2$ then $\angle C = 90^\circ$

$\angle C$ obtuse, AD ⊥ BC extended: $AB^2 = AC^2 + BC^2 + 2\,BC \cdot CD$

$\angle C$ acute: $AB^2 = AC^2 + BC^2 - 2\,BC \cdot CD$

Triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25

THEOREMS 36–38

36 **একই ভূমির উপর এবং একই সমান্তরাল রেখাযুগলের মধ্যে অবস্থিত সকল ত্রিভুজক্ষেত্রের ক্ষেত্রফল সমান।**

Complete each triangle into a parallelogram on the same base. Those parallelograms are equal, and each triangle is half of one.

37 **একই ভূমির উপর এবং একই সমান্তরাল রেখাযুগলের মধ্যে অবস্থিত ত্রিভুজক্ষেত্রের ক্ষেত্রফল সামান্তরিকক্ষেত্রের ক্ষেত্রফলের অর্ধেক।**

Draw a diagonal of the parallelogram: it makes a triangle on the same base, equal to the given one (Th 36), and that is half the parallelogram.

38 **সমকোণী ত্রিভুজের অতিভুজের উপর অঙ্কিত বর্গক্ষেত্রের ক্ষেত্রফল অপর দুই বাহুর উপর অঙ্কিত বর্গক্ষেত্রদ্বয়ের ক্ষেত্রফলের সমষ্টির সমান। (পিথাগোরাস)**

Book proof: draw squares on all 3 sides, join to make congruent triangles (SAS), each triangle = half a rectangle part. Quick check: two copies make a trapezium, ½(a+b)² = 2·½ab + ½c² ⇒ a² + b² = c².

Theorem numbers follow NCTB order (Pythagoras = 38, as you said). If your book's numbering is off by one somewhere, match by statement.

9 ত্রিকোণমিতিক অনুপাত Trig ratios

CQ SURE MCQ HEAVY

Ratios (angle θ, right triangle)

$\sin\theta = \frac{\text{লম্ব}}{\text{অতিভুজ}}$ $\cos\theta = \frac{\text{ভূমি}}{\text{অতিভুজ}}$ $\tan\theta = \frac{\text{লম্ব}}{\text{ভূমি}}$

$\operatorname{cosec}\theta = \frac{1}{\sin\theta}$ $\sec\theta = \frac{1}{\cos\theta}$ $\cot\theta = \frac{1}{\tan\theta}$

$\tan\theta = \frac{\sin\theta}{\cos\theta}$ $\cot\theta = \frac{\cos\theta}{\sin\theta}$

লম্ব = opposite side, ভূমি = adjacent side, অতিভুজ = hypotenuse.

Identities

$\sin^2\theta + \cos^2\theta = 1$

$\sec^2\theta - \tan^2\theta = 1 \Rightarrow (\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1$

$\operatorname{cosec}^2\theta - \cot^2\theta = 1$

$\sec^2\theta - 1 = \tan^2\theta$ $\operatorname{cosec}^2\theta - 1 = \cot^2\theta$

$\frac{1 - \sin^2\theta}{\sin^2\theta} = \cot^2\theta$

Value table (learn the sin row, the rest follows)

	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined
cot	undefined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	undefined
cosec	undefined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1

Trick: $\sin$ at $0, 30, 45, 60, 90 = \sqrt{\frac{0}{4}}, \sqrt{\frac{1}{4}}, \sqrt{\frac{2}{4}}, \sqrt{\frac{3}{4}}, \sqrt{\frac{4}{4}}$. cos is the same row reversed.

Complementary angles ($90^\circ - \theta$)

$\sin(90^\circ - \theta) = \cos\theta \quad \cos(90^\circ - \theta) = \sin\theta$

$\tan(90^\circ - \theta) = \cot\theta \quad \cot(90^\circ - \theta) = \tan\theta$

$\sec(90^\circ - \theta) = \operatorname{cosec}\theta \quad \operatorname{cosec}(90^\circ - \theta) = \sec\theta$

$\sin A = \cos A \Rightarrow A = 45^\circ$. $\sin 3A = \cos 3A \Rightarrow 3A = 45^\circ \Rightarrow A = 15^\circ$.

Extra (low chance, just in case)

$\sin(90^\circ + \theta) = \cos\theta$, $\cos(90^\circ + \theta) = -\sin\theta$, $\tan(90^\circ + \theta) = -\cot\theta$

$1^\circ = 60'$ (minute), $1' = 60''$ (second)

π radian $= 180^\circ$; $1^\circ = \frac{\pi}{180}$ rad; 1 rad $\approx 57.3^\circ$

None of your 340 MCQ slides used these. General Math Ch 9 is about acute angles 0° to 90°.

MCQ TRAPS (FROM YOUR SLIDES)

- $\sec\theta + \tan\theta = \frac{5}{2} \Rightarrow \sec\theta - \tan\theta = \frac{2}{5}$ (they multiply to 1). Then add/subtract to get each. $\sec A + \tan A = \sqrt{5} \Rightarrow \sin A = \frac{\tan A}{\sec A} = \frac{2}{3}$.
- $4\sin A = 3 \Rightarrow$ draw triangle: opposite 3, hyp 4, adjacent $\sqrt{7} \Rightarrow \tan A = \frac{3}{\sqrt{7}}$.
- $\sin\theta + \cos\theta = 1 \Rightarrow$ square $\Rightarrow \sin\theta\cos\theta = 0$.
- $\cot(A + 30^\circ) = 0 \Rightarrow A + 30^\circ = 90^\circ$. $\tan(60^\circ - \theta) = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$.
- $\cos^4\theta - \sin^4\theta = (\cos^2\theta - \sin^2\theta) \cdot 1$.

CQ PATTERN

(ক) given $\tan\theta = \frac{3}{4}$ (or similar), find others: draw a triangle, get the missing side by Pythagoras. (খ) prove identity: change everything to sin/cos, or multiply by the conjugate, e.g. $\sqrt{\frac{1-\sin A}{1+\sin A}} = \sec A - \tan A$. (গ) solve $0^\circ < \theta < 90^\circ$: e.g. $2\cos^2\theta + 3\sin\theta - 3 = 0 \Rightarrow$ use $\cos^2 = 1 - \sin^2 \Rightarrow (2\sin\theta - 1)(\sin\theta - 1) = 0 \Rightarrow \theta = 30^\circ$ (or 90° if allowed).

10 দূরত্ব ও উচ্চতা Height & distance

CQ SURE

TWO WORDS

উন্নতি কোণ (angle of elevation) = look **up** from ground. অবনতি কোণ (angle of depression) = look **down** from top. They are equal (alternate angles). Height and shadow equal $\Rightarrow$ angle 45°.

S1 · Tower from a point

Distance d, elevation $\theta \Rightarrow h = d\tan\theta$, $d = h\cot\theta$

S2 · Shadow

Shadow s, sun at $\theta \Rightarrow h = s\tan\theta$. Pole $6\sqrt{3}$ m, $\theta = 60^\circ \Rightarrow s = 6$ m

S3 · Ladder on wall

Length l, angle θ with ground $\Rightarrow$ height $= l\sin\theta$, foot from wall $= l\cos\theta$

20 m ladder, $30^\circ \Rightarrow$ height 10 m

S4 · Kite / string

String l, angle $\theta \Rightarrow$ height $= l\sin\theta$

S5 · Broken tree

Top touches ground at distance d from foot, angle θ with ground:

standing part $= d\tan\theta$, broken part $= \dfrac{d}{\cos\theta}$, full tree $=$ sum

S6 · From the top (depression)

Building h, depression θ to a point $\Rightarrow$ distance $= h\cot\theta = \dfrac{h}{\tan\theta}$

S7 · Two angles, walking closer (most common CQ (গ))

From far point angle β, walk a closer, angle becomes α ($\alpha > \beta$). Let near distance x:

$h = x\tan\alpha$ and $h = (x+a)\tan\beta \Rightarrow$ solve. Formula: $h = \dfrac{a\tan\alpha\tan\beta}{\tan\alpha - \tan\beta}$

Special: 30° then $60^\circ \Rightarrow x = \frac{a}{2}$, $h = \frac{\sqrt{3}}{2}a$. (Shadow grows a when sun goes $60^\circ \to 30^\circ$: same formula.)

S8 · Two buildings / opposite sides of road

Building height h_1. From its top: elevation α to tower top, depression β to tower foot. Gap $d = h_1\cot\beta$, tower $= h_1 + d\tan\alpha$.

Point between two posts on a road, angles α and β: road width $= h\cot\alpha + h\cot\beta$ (equal posts).

HOW TO ANSWER

(1) Draw the figure, mark right angle, angle, known side. (2) Pick the ratio that links known side and wanted side: $\tan$ for height/ground, $\sin$ or $\cos$ when the slanted side is involved. (3) Put $\sqrt{3} = 1.732$ only at the end. Write units (m).

16 পরিমিতি Mensuration

CQ · ONLY SKIMMED

Triangles

$\frac{1}{2}\times$ base $\times$ height

Two sides + angle: $\frac{1}{2}ab\sin C$ $(10, 12, 30^\circ \Rightarrow 30)$

Heron: $s = \frac{a+b+c}{2}$, area $= \sqrt{s(s-a)(s-b)(s-c)}$

Equilateral: area $\frac{\sqrt{3}}{4}a^2$, height $\frac{\sqrt{3}}{2}a$ (= median)

Isosceles (base a, equal sides b): $\frac{a}{4}\sqrt{4b^2 - a^2}$

Right: $\frac{1}{2}\times$ two legs. Height to hypotenuse $= \frac{\text{leg}_1 \times \text{leg}_2}{\text{hyp}}$

Quadrilaterals

Rectangle: lb, perimeter $2(l+b)$, diagonal $\sqrt{l^2+b^2}$

Square: $a^2 = \frac{d^2}{2}$, diagonal $a\sqrt{2}$, perimeter $4a$

Parallelogram: base $\times$ height

Rhombus: $\frac{1}{2}d_1d_2$, side $= \frac{1}{2}\sqrt{d_1^2 + d_2^2}$

Trapezium: $\frac{1}{2}(a+b)\times h$

Regular n-gon (side a): $\frac{na^2}{4}\cot\frac{180^\circ}{n}$

Circle

Circumference $2\pi r$, area πr^2

Arc length $s = \dfrac{\theta}{360^\circ} \times 2\pi r = \dfrac{\pi r \theta}{180^\circ}$

Sector area $= \dfrac{\theta}{360^\circ} \times \pi r^2$

Ring (between circles): $\pi(R^2 - r^2)$

Wheel turns $= \dfrac{\text{distance}}{2\pi r}$

Square's incircle radius $= \frac{\text{side}}{2}$

Solids

Box: volume lbh, surface $2(lb + bh + hl)$, diagonal $\sqrt{l^2 + b^2 + h^2}$

Cube: volume a^3, surface $6a^2$, diagonal $\sqrt{3}\,a$

Cylinder: volume $\pi r^2 h$, curved $2\pi rh$, total $2\pi r(r + h)$

4 walls of a room: $2h(l + b)$

$1\ \text{m}^3 = 1000\ \text{L},\ \ 1\ \text{L} = 1000\ \text{cm}^3$

THE "CHANGE" PROBLEMS (HOW TO SOLVE EACH)

1 · Sides grow, area grows by a given amount

Equilateral side $+2$ cm, area $+6\sqrt{3}$. Old side a:

$\frac{\sqrt{3}}{4}\{(a+2)^2 - a^2\} = 6\sqrt{3} \Rightarrow \frac{\sqrt{3}}{4}(4a + 4) = 6\sqrt{3} \Rightarrow a + 1 = 6 \Rightarrow a = 5$

Rule: write NEW area − OLD area = given change. The a^2 cancels, leaving a linear equation.

2 · Percent change

Side $+x\%$ ⇒ area $+\left(2x + \frac{x^2}{100}\right)\%$. $10\% \Rightarrow 21\%$

Length $+x\%$, width $-y\%$ ⇒ $\left(x - y - \frac{xy}{100}\right)\%$

Side $\times k$ ⇒ area $\times k^2$, volume $\times k^3$

Radius doubled ⇒ circumference ×2, area ×4

3 · Right triangle: perimeter + hypotenuse

Hyp 13, perimeter 30 ⇒ $a + b = 17$, $a^2 + b^2 = 169$

$2ab = (a+b)^2 - (a^2 + b^2) = 120 \Rightarrow$ area $= \frac{ab}{2} = 30$

4 · Sides in a ratio

Sides $3 : 4 : 5$, perimeter 36 ⇒ $12k = 36$, $k = 3 \Rightarrow 9, 12, 15$ ⇒ area 54

Isosceles: perimeter 16, base 6 ⇒ sides 5, 5 ⇒ $\frac{6}{4}\sqrt{100 - 36} = 12$

5 · Path around a garden

Outside path width w: $(l + 2w)(b + 2w) - lb$

Inside path: $lb - (l - 2w)(b - 2w)$

Two crossing paths in middle: $w(l + b) - w^2$

40×30 garden, 2 m path outside ⇒ $44 \times 34 - 1200 = 296\ \text{m}^2$

6 · Wooden box (thickness t)

Closed box: inner $= (l - 2t)(b - 2t)(h - 2t)$

Open top: inner height $= h - t$

Wood = outer volume − inner volume

$10 \times 8 \times 6$, $t = 1$, closed ⇒ $480 - 192 = 288\ \text{cm}^3$

7 · Rectangle from area + perimeter

$l + b = \frac{P}{2}$, $lb = A \Rightarrow$ diagonal $= \sqrt{(l+b)^2 - 2lb}$

Length 3 more than width, perimeter 30 ⇒ $w = 6$

8 · Cost / tiles / melting

Cost = area × rate. Tiles $= \frac{\text{floor area}}{\text{tile area}}$

Melt and recast: volume stays the same

Painting a room: $2h(l + b)$ (+ lb for ceiling) − doors/windows

17 পরিসংখ্যান Statistics

CQ SURE EASY MARKS

Setup

পরিসর (range) $= (\max - \min) + 1$ (book rule, +1)

Number of classes $= \dfrac{\text{range}}{\text{class width}}$ (round **up**)

Class midpoint $= \dfrac{\text{lower} + \text{upper}}{2}$

ক্রমযোজিত গণসংখ্যা (cumulative frequency): keep adding f

বিচ্ছিন্ন (discrete, counted): population, students, year. অবিচ্ছিন্ন (continuous, measured): age, height, weight.

Mean (গড়)

Raw: $\bar{x} = \dfrac{\sum x}{n}$

Grouped: $\bar{x} = \dfrac{\sum f_i x_i}{n}$ (x_i = midpoint)

সংক্ষিপ্ত পদ্ধতি: $u_i = \dfrac{x_i - a}{h}$

$\bar{x} = a + \dfrac{\sum f_i u_i}{n} \times h$

a = assumed mean (a middle midpoint), h = class width. $a = 16$, mean of $u = 1$, $h = 4 \Rightarrow 20$

Median (মধ্যক)

Raw, sort first. n odd: $\left(\frac{n+1}{2}\right)$th value

n even: average of $\frac{n}{2}$th and $\left(\frac{n}{2}+1\right)$th

Grouped: $\text{Me} = L + \left(\dfrac{n}{2} - F_c\right) \times \dfrac{h}{f_m}$

L = lower limit of median class (first class where cumulative $f \geq \frac{n}{2}$), F_c = cumulative f before it, f_m = its frequency.

Mode (প্রচুরক)

Raw: the most repeated value.

Grouped: $\text{Mo} = L + \dfrac{f_1}{f_1 + f_2} \times h$

$f_1 = f_{\text{mode}} - f_{\text{before}}$, $f_2 = f_{\text{mode}} - f_{\text{after}}$

Classes 31–40: 16, 41–50: 20, 51–60: 14 $\Rightarrow f_1 = 4$, $f_2 = 6 \Rightarrow 41 + \frac{4}{10} \times 10 = 45$. (Your slide's 47 used f_2 on top: wrong.)

Graphs (CQ (গ) usually asks one)

আয়তলেখ (histogram): classes must be **continuous**. Make boundaries: 31–40, 41–50 ⇒ 30.5–40.5, 40.5–50.5. Bars touch.

গণসংখ্যা বহুভুজ (frequency polygon): join the tops' midpoints of histogram bars, close at both ends to the axis.

অজিভ রেখা (ogive): x = upper boundary, y = **cumulative frequency**. Smooth curve.

পাই চিত্র: angle $= \dfrac{f}{n} \times 360^\circ$

Always write the scale: "x-axis 1 square = 5 units".

MCQ TRAPS

- Min 35, range 56 ⇒ max $= 56 + 35 - 1 = 90$.
- Median of $2, 10, 5, 1, 0, 7, 3, 4$: sort first ⇒ $0, 1, 2, 3, 4, 5, 7, 10 \Rightarrow \frac{3+4}{2} = 3.5$.
- Primes 12 to 45: 13, 17, 19, 23, **29**, 31, 37, 41, 43 ⇒ median 29.

! Slide answer check where your annotated slides look wrong

Ch 1: "Two irrationals: sum always irrational, difference always irrational, product can be either". Slide marks all three. Correct: **only iii**.

Ch 2: "$A \cap B = \varnothing \Rightarrow$ i. disjoint ii. finite iii. infinite". Slide marks all three. Correct: **only i**.

Ch 17: grouped mode slide (21–30: 6, 31–40: 16, 41–50: 20, 51–60: 14). Slide gives 47. Correct: **45**.

Ch 14: scalene triangle lines of symmetry = **0** (no option on slide fits).

If the real exam has the same question, go with the math, not the slide mark.

⚡ 20 min Mobile Sprint

For the commute. Read top to bottom. Each block = the highest-value stuff only. Also sent as a separate phone-size PDF.

0–3 min Algebra formulas

$a^2+b^2=(a+b)^2-2ab$ · $(a-b)^2=(a+b)^2-4ab$ · $a^3+b^3=(a+b)^3-3ab(a+b)$ · $x^3+\frac{1}{x^3}=(x+\frac{1}{x})^3-3(x+\frac{1}{x})$ · $a+b+c=0\Rightarrow a^3+b^3+c^3=3abc$ · remainder $=f(a)$ · $n(P(A))=2^n$, proper 2^n-1 · $n(A\cup B)=n(A)+n(B)-n(A\cap B)$

3–6 min Log, equations, ratio, series

$\log_a MN=\log M+\log N$ · $\log M^r=r\log M$ · $\log_a a=1$, $\log_a 1=0$ · base $>0,\neq 1$ · characteristic = power of 10 in $a\times 10^n$ · never divide by x ($x^2=3x\Rightarrow\{0,3\}$) · check roots in $\sqrt{}$ and denominators · যোজন-বিয়োজন $\frac{a+b}{a-b}=\frac{c+d}{c-d}$ · $b^2=ac$ · AP $t_n=a+(n-1)d$, $S_n=\frac{n}{2}\{2a+(n-1)d\}$ · GP $t_n=ar^{n-1}$ · $\sum n^3=\{\frac{n(n+1)}{2}\}^2$

6–9 min Trig

sin: $0,\frac{1}{2},\frac{1}{\sqrt{2}},\frac{\sqrt{3}}{2},1$ · cos reversed · $\tan 30=\frac{1}{\sqrt{3}}$, $\tan 60=\sqrt{3}$ · $\sin^2+\cos^2=1$ · $(\sec+\tan)(\sec-\tan)=1$ · $\sin(90-\theta)=\cos\theta$ · height $=d\tan\theta$ · ladder height $=l\sin\theta$ · $30^\circ\to 60^\circ$ walking $a\Rightarrow h=\frac{\sqrt{3}}{2}a$

9–13 min Geometry

bisectors of B, C $\Rightarrow\angle BOC=90+\frac{A}{2}$ · midpoint line $=\frac{1}{2}$ base · inscribed $=\frac{1}{2}$ central · semicircle 90° · cyclic opposite sum 180° · tangent $\perp$ radius · 2 tangents from outside are equal · tangents: sep 4, touch-out 3, cut 2, touch-in 1 · touch: $d=r_1\pm r_2$ · similar area ratio = side ratio^2 · bisector $\frac{BD}{DC}=\frac{AB}{AC}$ · symmetry: square 4, rhombus 2, equilateral 3, scalene 0, circle infinite · rotation angle $=\frac{360}{\text{order}}$ · diagonal halves a parallelogram

13–17 min Mensuration + stats

equilateral $\frac{\sqrt{3}}{4}a^2$ · Heron · $\frac{1}{2}ab\sin C$ · rhombus $\frac{1}{2}d_1d_2$ · trapezium $\frac{1}{2}(a+b)h$ · arc $\frac{\pi r\theta}{180}$ · cylinder πr^2h · box diagonal $\sqrt{l^2+b^2+h^2}$ · side $+x\%\Rightarrow$ area $+(2x+\frac{x^2}{100})\%$ · range = max − min + 1 · $\bar{x}=a+\frac{\sum fu}{n}h$ · Me $=L+(\frac{n}{2}-F_c)\frac{h}{f_m}$ · Mo $=L+\frac{f_1}{f_1+f_2}h$ · ogive y-axis = cumulative frequency · histogram needs continuous classes

17–20 min Exam moves

1. MCQ first, 30 min. Unsure? Eliminate 2, then pick. Don't leave blanks.
2. CQ: pick one from each group first (Algebra, Geometry, Trig/Mensuration, Statistics), then your strongest.
3. Theorem: সাধারণ নির্বচন → চিত্র → বিশেষ নির্বচন → অঙ্কন → প্রমাণ.
4. Every (ক) is 2 easy marks. Write it even if you skip (গ).
5. Last 10 min: check signs, units, and that you answered what was asked.